

$$L = -[\gamma \log(a) + (1-\gamma) \log(1-a)]$$

$$\frac{dL}{d\cancel{a}} = -\left[\frac{\gamma}{a} + (1-\gamma) \cdot \frac{1}{1-a} \frac{d}{d\cancel{a}}(1-a)\right]$$

$$= -\left[\frac{\gamma}{a} + \frac{1-\gamma}{1-a} (0-1)\right]$$

$$= -\left[\frac{\gamma}{a} - \frac{1-\gamma}{1-a}\right]$$

$$= \frac{-(1-a)\gamma + a(1-\gamma)}{a(1-a)}$$

$$= \frac{-\gamma + a\gamma + a - a\gamma}{a(1-a)}$$

$$= \frac{a-\gamma}{a(1-a)}$$